

NEPTUN-code:, Name:

Group, teacher:

Test score:

Computer Science BSc, Analysis I, TEST-2

13th of May, 2025

Reasoning and justification are needed in the solutions.

1. (5 points) Evaluate the following sum:

$$\sum_{n=1}^{+\infty} \frac{2^{-1} \cdot 3^{n+2} + (-1)^{2n+1}}{5^n}.$$

2. (6+6+6 points) Determine whether the following series are convergent or divergent. Prove your statements.

$$\text{a) } \sum_{n=0} \frac{n^2 - n + 7}{n^4 + \sqrt{n^4 + 1}}, \quad \text{b) } \sum_{n=0} \left(\frac{2n-3}{2n+1} \right)^{n^2+2n}, \quad \text{c) } \sum_{n=0} \frac{(n+2)!}{(n+1)^2 \cdot 5^{n+2}}.$$

3. (8 points) Determine the convergence set of the following power series:

$$\sum_{n=0} \frac{3^n}{n+1} \cdot (x+1)^n \quad (x \in \mathbb{R}).$$

4. (7 points) Evaluate the following limit, and prove by definition your result:

$$\lim_{x \rightarrow -2} \frac{x^2 - 4}{x^3 - x^2 - 6x}$$

5. (6+6 points) Evaluate the following limits (do not use L'Hospital's rule here):

$$\text{a) } \lim_{x \rightarrow 0} \frac{e^{3x} - e^x}{\sqrt{x^2 + x + 3} - \sqrt{3 - 3x}}, \quad \text{b) } \lim_{x \rightarrow 0} \frac{x(1 - \cos x)}{x - \sin x}.$$